

Photosynthesis — Complete Notes

Grade 11

Life Sciences

CAPS

What you'll learn

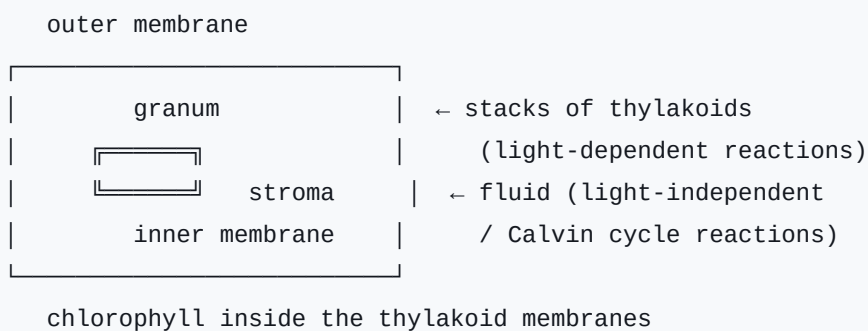
- What photosynthesis is and why it is essential for life on Earth
- The structure of a chloroplast and how it relates to its function
- The requirements for photosynthesis and the word and balanced chemical equations
- The two phases: the light-dependent reactions and the light-independent (Calvin cycle) reactions
- The factors that affect the rate of photosynthesis, and how they interact
- How to investigate photosynthesis experimentally
- How to answer full process and data questions

What is photosynthesis?

Photosynthesis is the process by which green plants, algae and some bacteria use **light energy** to convert **carbon dioxide** and **water** into **glucose** (a carbohydrate) and **oxygen**.

- Organisms that make their own food in this way are called **autotrophs** (producers).
- The light energy is captured by the green pigment **chlorophyll**, which is found in **chloroplasts**.
- The energy is stored as **chemical potential energy** in the glucose that is made.
- Photosynthesis is the **entry point** of energy into almost every food chain, and it releases the **oxygen** that most living things need for respiration.

Where it happens: the chloroplast



- **Thylakoids** are flattened discs stacked into **grana**; the **chlorophyll** in their membranes absorbs light energy. This is where the **light-dependent** reactions occur.
- **Stroma** is the fluid around the grana; this is where the **light-independent** (Calvin cycle) reactions occur.

Requirements for photosynthesis

Requirement	Role
Light (sunlight)	Provides the energy
Carbon dioxide (CO ₂)	Source of carbon for glucose
Water (H ₂ O)	Source of hydrogen and oxygen
Chlorophyll	Absorbs light energy
Enzymes	Control the reactions (in the chloroplast)
Suitable temperature	Enzymes work best at an optimum

The equations

Word equation:

carbon dioxide + water → glucose + oxygen
(light energy and chlorophyll are needed)

Balanced chemical equation:

$$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$$

Photosynthesis and respiration are **opposite** processes — what respiration uses, photosynthesis makes, and vice versa.

The two phases of photosynthesis

Phase 1: The light-dependent reactions (in the thylakoids)

- **Light energy** is absorbed by chlorophyll.
- **Water is split** (this is called **photolysis**): $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$.
- The **oxygen** is released as a by-product.
- The energy from light is used to make **ATP** and **NADPH** (reduced NADP), which carry energy to the next phase.

Phase 2: The light-independent reactions (in the stroma) — the Calvin cycle

- **Carbon dioxide** is **fixed** — bonded into an organic molecule.
- The energy carried by **ATP** and **NADPH** from phase 1 is used to convert the carbon into **glucose**.
- These reactions do **not** need light directly, but they cannot continue for long without the products of the light phase.
- They are sometimes called the **dark reactions**, although they do not occur only in darkness.

Factors affecting the rate of photosynthesis

- **Light intensity**: as light intensity increases, so does the rate, up to a point. Beyond that point the rate **plateaus** because another factor (usually CO₂ or temperature) becomes **limiting**.

- **Carbon dioxide concentration:** increasing CO₂ increases the rate, until light or temperature becomes limiting.
- **Temperature:** the rate increases with temperature up to an **optimum** (about 25–35 °C for many plants); above this the **enzymes denature** and the rate falls sharply.
- **Water:** too little water closes the stomata, reducing CO₂ intake, so the rate drops.
- **Chlorophyll:** leaves without chlorophyll (e.g. the white patches of a variegated leaf) cannot photosynthesise there.

Any factor that is in short supply and slows the rate is called a **limiting factor**.

Investigating photosynthesis

1. Destarching

Before any starch test, the plant is kept in the **dark for 24–48 hours**. This lets it use up its stored starch, so any starch found afterwards must have been made during the experiment.

2. Testing a leaf for starch (iodine test)

1. A leaf is boiled in water (to kill the cells and break the membranes), then boiled in ethanol (to remove chlorophyll so the colours are clear).
2. The leaf is rinsed and covered with **iodine solution**.
3. A **blue-black** colour shows that **starch** was made; a brown colour means no starch.

This test shows **whether** photosynthesis happened.

3. Testing that chlorophyll is needed (variegated leaf)

A variegated leaf has green and white parts. After the iodine test, **only the green parts go blue-black**, because only they contain **chlorophyll**. This proves that chlorophyll is needed.

4. Testing that light is needed

Part of a leaf is covered with foil (or the plant is kept in the dark). After the test, only the exposed part turns blue-black. This shows that **light** is needed.

5. Testing that CO₂ is needed

A plant is enclosed with **soda lime** (which absorbs CO₂). After the iodine test, the leaf makes no starch when CO₂ is absent.

6. Testing for the oxygen released (bubbles)

An aquatic plant such as ***Elodea*** in water and light releases gas bubbles. If the gas is collected and a **glowing splint** is held in it, the splint relights — proof that the gas is **oxygen**.

Worked examples

Worked example 1 — Explain the process

Explain how the products of the light-dependent reactions are used in the light-independent reactions.

Step 1 — In the light-dependent reactions (in the thylakoid membranes), light energy splits water (**photolysis**) and produces **ATP**, **NADPH** and **oxygen**.

Step 2 — The ATP and NADPH move into the **stroma**.

Step 3 — In the stroma, **CO₂ is fixed** into an organic molecule (via the Calvin cycle).

Step 4 — The energy in **ATP** and the reducing power of **NADPH** are used to build this into **glucose**.

Step 5 — The oxygen from photolysis is released as a by-product (it is not used by the plant for this process, though it is used in respiration).

Answer: ATP supplies energy and NADPH supplies hydrogen and electrons to fix CO₂ into glucose in the stroma.

Worked example 2 — Interpret a graph

A graph shows the rate of photosynthesis (y-axis) rising steeply as light intensity (x-axis) increases, then levelling off into a flat plateau. Explain the shape.

Step 1 — The rising section. At first **light intensity is the limiting factor**; more light means more energy captured, so the rate increases.

Step 2 — The plateau section. Once the line flattens, light is no longer limiting; some **other factor** — usually CO₂ concentration or temperature — has become the **limiting factor**, so adding more light cannot increase the rate.

Step 3 — Prediction. If CO₂ were increased at that point, the plateau would rise.

Answer: The curve rises while light limits the rate, then plateaus when another factor becomes limiting.

Worked example 3 — Design and reason (calculation)

A learner counts 30 bubbles released by an *Elodea* plant in one minute at a light intensity of 20 cm from the lamp, and 45 bubbles at 10 cm. Calculate the percentage increase.

Step 1 — Increase = 45 – 30 = 15 bubbles.

Step 2 — Percentage increase = (increase ÷ original) × 100 = (15 ÷ 30) × 100 = **50%**.

Step 3 — Reason. Moving the lamp closer increased light intensity, which increased the rate of photosynthesis and therefore the rate of oxygen release.

Answer: 50% increase.

Common mistakes

- Saying plants photosynthesise **instead of** respiring. Green plants respire **all the time**; they also photosynthesise in the light.
- Confusing the two phases: the light-dependent phase happens in the **thylakoids** and releases **oxygen**; the light-independent (Calvin) phase happens in the **stroma** and makes **glucose**.
- Writing the word equation the wrong way round, or forgetting that "light energy and chlorophyll are needed".
- Thinking the "dark reactions" happen only at night — they do **not** need light directly, but they depend on the products of the light phase.
- Saying the rate keeps rising forever with more light — it **plateaus** when another factor becomes limiting.
- Skipping the **destarching** step and then wrongly concluding that the starch came from the experiment.
- Forgetting that above the **optimum temperature** the enzymes **denature**, so the rate drops.

Quick summary / cheat sheet

- **Equation:** $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (light and chlorophyll needed).
- **Light phase (thylakoids):** light splits water $\rightarrow \text{ATP} + \text{NADPH} + \text{O}_2$.
- **Calvin cycle (stroma):** $\text{ATP} + \text{NADPH}$ fix $\text{CO}_2 \rightarrow$ glucose.
- **Limiting factors:** light intensity, CO_2 concentration, temperature, water, chlorophyll.
- **Tests:** iodine (blue-black = starch), variegated leaf (chlorophyll), foil (light), soda lime (CO_2), *Elodea* with a glowing splint (oxygen).
- **Always destarch first.**

Practice questions

1. Write the balanced chemical equation for photosynthesis in words and in symbols.
2. State **two** requirements for photosynthesis and give the role of each.
3. Where in the chloroplast do the light-dependent and the light-independent reactions take place?
4. Explain why the rate of photosynthesis stops increasing at very high light intensities.
5. Why must a plant be destarched before a starch (iodine) test?
6. A plant is kept at 45 °C. Explain why its rate of photosynthesis is very low even though light and CO_2 are plentiful.
7. An *Elodea* plant releases 24 bubbles in 2 minutes. Calculate the rate per minute.

Answers

1. Words: carbon dioxide + water $\rightarrow$ glucose + oxygen (light energy and chlorophyll needed). Symbols:
 $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
2. Any two of: light (provides energy); CO_2 (source of carbon); water (source of hydrogen and oxygen); chlorophyll (absorbs light); enzymes (control the reactions).
3. Light-dependent $\rightarrow$ **thylakoid membranes (grana)**; light-independent (Calvin cycle) $\rightarrow$ **stroma**.
4. At high light intensity, light is no longer the limiting factor; **another factor** such as CO_2 concentration or temperature becomes limiting, so the rate plateaus even if the light increases.
5. Destarching removes the plant's **existing stored starch**, so any starch detected after the experiment must have been produced **during** it (a fair test).
6. At 45 °C the temperature is above the optimum, so the **enzymes** that control photosynthesis are **denatured** and the reactions cannot proceed.
7. Rate = $24 \div 2 =$ **12 bubbles per minute**.

Official EduRank Team notes. Always cross-check with your textbook and teacher.